Assignment No-01

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Course : CSE350

Section : 15

Ans to the Q No-1

(a) Here, $V_{ref} = -10V$, $R_f = 10k\Omega$, $R = 5k\Omega$.

$$I_1 = \frac{V_{ref}}{2R} = \frac{-10}{10} mA = -1 mA.$$

$$I_2 = \frac{I_1}{2} = \frac{-1}{2} mA = -0.5 mA.$$

$$I_3 = \frac{I_2}{2} = \frac{-0.5}{2} mA = -0.25 mA.$$

(Ans)

(b) Resolution or 1 LSB value,

$$V_o = -V_{ref} \times \frac{R_f}{2R}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4}\right)$$

$$= -(-10) \times \frac{10}{10}\left(0 + 0 + \frac{1}{4}\right)$$

$$= 2.5 V.$$

Ans

(c) Maximum value of $V_o = -V_{ref} \times \frac{R_f}{2R}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4}\right)$

$$= -(-10) \times \frac{10}{10}\left(1 + \frac{1}{2} + \frac{1}{4}\right)$$

$$= 17.5 V.$$

Minimum value of $V_o = -V_{ref} \times \frac{R_f}{2R}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4}\right)$

$= -(-10) \times \frac{10}{10} (0+0+0)$

$= 0V.$

Ans

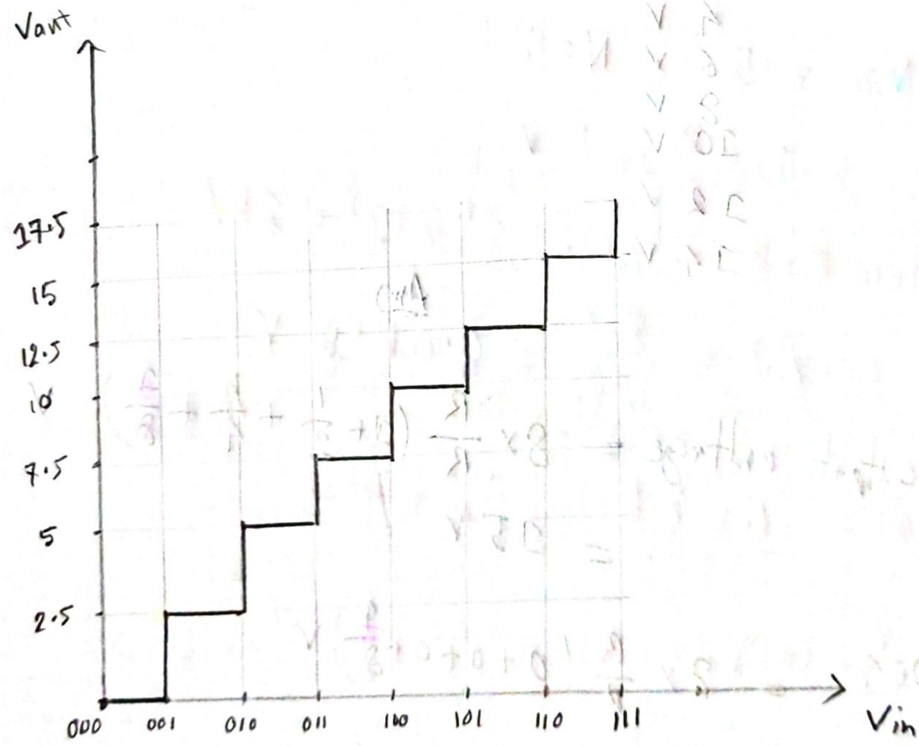
(d)

$\frac{1}{1}$ MSB value $= -(-10) \times \frac{10}{10} (1+0+0) = 10V.$

$\therefore \frac{1}{2}$ MSB value $= \frac{10}{2} V = 5 V.$

Ans

(e)



Ans to the Q No-2

(a) $V_R = -8V$, $R_3 = R_f$

so, For XXX0 where X canbe 0 or 1,

$$V_0 = V_{ref} \times \frac{R_f}{R}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8}\right)$$

$$\therefore V_0 = V_{ref}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4}\right) \quad [\because R_f = R \text{ and } b_4 = 0]$$

XXX0	output voltage
0000	0V
0010	2 V
0100	4 V
0110	6 V
1000	8 V
1010	10 V
1100	12 V
1110	14 V.

Ans

(b)

$$\text{Maximum output voltage} = 8 \times \frac{R}{R}\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right) V$$

$$= 15 V.$$

$$\text{LSB or } 0001 = 8 \times \frac{R}{R}\left(0 + 0 + 0 + \frac{1}{8}\right) V$$

$$= 1 V.$$

$$\text{Ratio} = \frac{15}{1}$$

ratio of maximum output voltage and LSB is = 15:1.

Ans

(c) As $R_3 = R_f$ and it is a BWR DAC,

$$V_o = R_f \times \left(b_1 \times \frac{V_{ref}}{R_3} + b_2 \times \frac{V_{ref}}{2R_3} + b_3 \times \frac{V_{ref}}{4R_3} + b_4 \times \frac{V_{ref}}{8R_3}\right)$$

$$\Rightarrow V_o = V_{ref} \times \frac{R_f}{R_3}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8}\right)$$

$$\Rightarrow \frac{R_3}{R_f} = \frac{V_{ref}}{V_o}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8}\right)$$

$$\Rightarrow \frac{R_3}{R_f} = \frac{8}{5}\left(1 + 0 + \frac{1}{4} + 0\right) \quad \left[\because V_o = 5V,\ V_{ref} = 8V \text{ and input} = 1010\right]$$

$$\Rightarrow \frac{R_3}{R_f} = 2$$

Ratio of R_3 and R_f will be $= 2:1$

Ans to the Q No - 3

(a) Here, $V_{ref} = 10V$, $R_1 = 5k\Omega$, $R_2 = 10k\Omega$, $R_3 = 10k\Omega$, $R_4 = 15k\Omega$.

Input is $0 - 10V$,

$V_1 = 0V$

$$V_2 = \frac{5}{(5+10+10+15)}(10-0) + 0$$

$= 1.25V$.

$$V_3 = \frac{15}{40}(10) = 3.75V$$

$$V_4 = \frac{25}{40} \times 10 = 6.25V.$$

Quantization Range	Digital output
6.25 - 10	11
3.75 - 6.25	10
1.25 - 3.75	01
0 - 1.25	00

(b) Using the table from 'a' and sample value from table 1

Table from 'a'		Identification of sample value		
Quantization Range	Digital output	Sampled value	Sampled values quantization range	Sampled values encoded output
6.25-10	11	5	3.75-6.25	10
3.75-6.25	10	7.4	6.25-10	11
1.25-3.75	01	5.71	3.75-6.25	10
0-1.25	00	1.22	0-1.25	00

Ans to the Q No-4

(a) Here $V_{ref} = 5V$, $N = 8$ bit.

voltage range is 0 to 5 V.

~~steps~~ number of steps $= 2^N = 2^8 = 256$

step size of ADC $= \frac{5-0}{256} = 0.0195$ V.

Ans

(b) $n = 200$, $N = 8$ bit, $V_{ref} = 5V$.

$V_{input} = \frac{n}{2^N} \times V_{ref}$

$= \frac{200}{2^8} \times 5$

$= 3.90$ V.

Ans

(c) $N = 8$ bit, $n = 200$, $f = 10^6$ Hz so $T = \frac{1}{10^6}$ s

$\therefore t_1 = 2^N T = 2^8 \times \frac{1}{10^6}$ s $= 2.56 \times 10^{-4}$ s

$\therefore t_2 = nT = 200 \times \frac{1}{10^6}$ s $= 2 \times 10^{-4}$ s.

$\therefore$ Total reading time $= (t_1 + t_2)$

$= (2.56 \times 10^{-4} + 2 \times 10^{-4})$ s

$= 4.56 \times 10^{-4}$ s

(Ans)

(d) max sample rate is when $t_1 = t_2$.

$\therefore$ maximum sample rate $= \frac{1}{t_1 + t_2} = \frac{1}{2 \times (2^N T)} = \frac{10^6}{2 \times 2^8} = 1953.12$ Hz

(e) sample rate $= \frac{1}{t_1 + t_2}$.

$\therefore$ sample rate $\propto \frac{1}{2^N}$ assuming others parameters ~~as contents~~. constants

So if we increase N by 1 sample rate will decrease by 2. So if we wish to increase sample rate by 4 times we have to decrease N by 2 times.

$\therefore$ counter bits will be $= (8 - 2)$ bit

$= 6$ bit

Ans